BTS-III(S)-(11.20/04.21)-1697 Reg. No. | | | | | | | | |

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B.Tech. Degree III Semester Supplementary Examination November 2020/April 2021

ME 15-1303 MECHANICS OF SOLIDS
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) State and explain Hooke's law.

(b) Differentiate between plain stress and plane strain.

(c) Explain the significance of Mohr's circle and its uses.

(d) List the assumptions in torsion equation.

(e) What is point of contra flexure? Illustrate with an example.

(f) Draw the shear force and bending moment diagrams for a cantilever of length L carrying a uniformly distributed load of w per meter length over its entire length.

(g) What are the assumptions made in theory of simple bending?

(h) Sketch and explain the shear stress distribution in a symmetrical I section.

(i) What is Macaulay's method? Where is it used?

(j) What is meant by equivalent length of a column? What are its values under typical loading conditions?

PART B

$(4 \times 10 = 40)$

II. Two vertical rods one of steel and other of copper are each rigidly fixed at the top and 50 cm apart. Diameters and lengths of each rod are 2 cm and 4 m respectively. A cross bar fixed to the rods at the lower ends carries a load of 5000 N such that the cross bar remains horizontal even after loading. Find the stress in each rod and the position of the load on the bar. Take E for steel = 2×10^5 N/mm^2 and E for copper = 1×10^5 N/mm^2. (10)

OR

III. The normal stresses in two mutually perpendicular directions are 600 N/mm^2 and 300 N/mm^2 both tensile. The complimentary shear stresses in these directions are of intensity 450 N/mm^2. Find the normal and tangential stresses on the two planes which are equally inclined to the planes carrying the normal stresses mentioned above. (10)

(P.T.O)

IV. Determine the diameter of a solid shaft which will transmit 300 kW at 250 r.p.m. The maximum shear stress should not exceed 30 N/mm^2 and twist should not be more than 1° in a shaft of length 2 m. Take modulus of rigidity = 1×10^5 N/mm^2. (10)

OR

V. Draw the shear force and bending moment diagram for a simply supported beam of length 9 m and carrying a uniformly distributed load of 10 kN/m for a distance of 6 m from the left end. Also calculate the maximum bending moment on the session. (10)

VI. A beam of rectangular section of length 8 m is simply supported. The beam carries a uniformly distributed load of 12 kN/m run over the entire length and a point load of 10 kN at 3 meter from the left support. If the depth is two times the width and the stress in the material is not to exceed 8 N/mm^2, find the suitable dimensions of the section. (10)

OR

VII. Discuss on various theories of failures and their applications to ductile and brittle materials. (10)

VIII. A beam of length 6 m is simply supported at its ends and carries two point loads of 48 kN and 40 kN at a distance of 1 m and 3 m respectively from the left support. Find the deflection under each load and also the maximum deflection. Take $E = 2 \times 10^5$ N/mm^2 and $I = 85 \times 10^6$ mm^4. (10)

OR

IX. A hollow cast iron column whose outside diameter is 200 mm has a thickness of 20 mm. It is 4.5 m long and is fixed at both ends. Calculate the safe load by Rankine's formula using a factor of safety of 4. Calculate the slenderness ratio and the ratio of Euler's and Rankine's critical loads. Take crushing stress = 550 N/mm^2, a = 1/1600 in Rankine's formula and $E = 9.4 \times 10^4$ N/mm^2. (10)

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